IDEMPOTENT ENDOMORPHISMS OF THE OUTER TERMS EXTEND TO THE TRIANGLE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 746-755, following the statement at 730-744.

Editorial note. The polynomial identity check is substitution-level algebra and is flagged as such in the notes.

Allowed standard prerequisites (editorial).

- Square-zero endomorphisms of the middle term. lemma-third-map-square-zero (carded L26251, cited at ~745). (b')^2 - b' has square zero.

Source reference. Original lemma 05QP: sources/stacks-derived/pinned-original-source.tex, source lines 701–701. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a pre-triangulated category. Let (X, Y, Z, f, g, h) be a distinguished triangle. If*

$$\begin{array}{ccc} Z & \xrightarrow[h]{} & X[1] \\ {\scriptstyle c}\downarrow & & \downarrow{\scriptstyle a[1]} \\ Z & \xrightarrow{h} & X[1] \end{array}$$

is commutative and $a^2 = a$, $c^2 = c$, then there exists a morphism $b : Y \to Y$ with $b^2 = b$ such that (a, b, c) is an endomorphism of the triangle (X, Y, Z, f, g, h).

Proof. By TR3 there exists a morphism b' such that (a, b', c) is an endomorphism of (X, Y, Z, f, g, h). Then $(0, (b')^2 - b', 0)$ is also an endomorphism. By Lemma 05QP we see that $(b')^2 - b'$ has square zero. Set $b = b' - (2b' - 1)((b')^2 - b') = 3(b')^2 - 2(b')^3$. A computation shows that (a, b, c) is an endomorphism and that $b^2 - b = (4(b')^2 - 4b' - 3)((b')^2 - b')^2 = 0$. □

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.